S4. Appendix 4. Harvest Plot

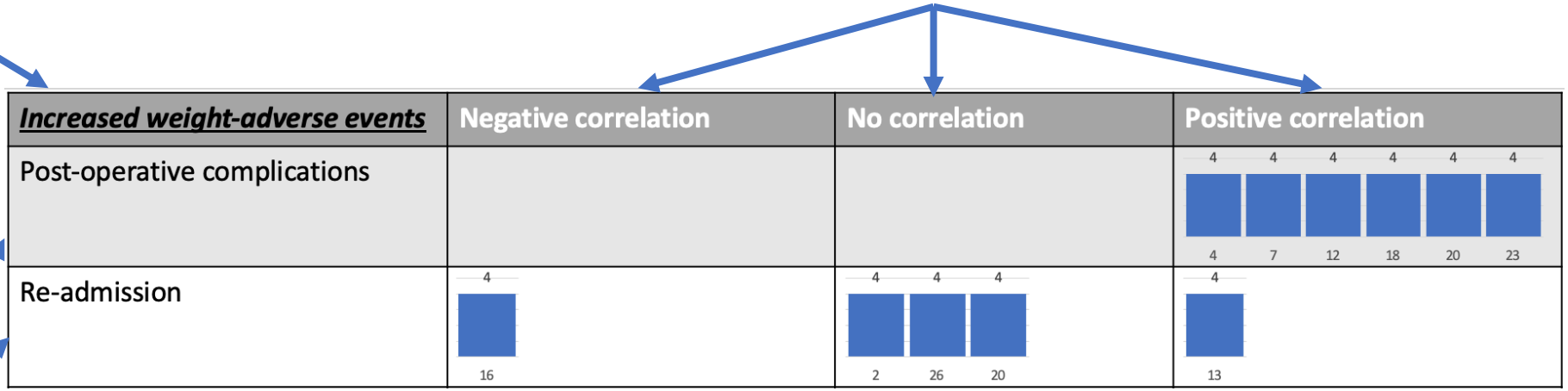
This is a visual representation of the specific outcomes assessed in individual papers (refer to S5.Appendix 5 for bibliography).

Understanding the harvest plot:

The information is displayed in the following format:

The variable being examined

The relationship between variable and outcome

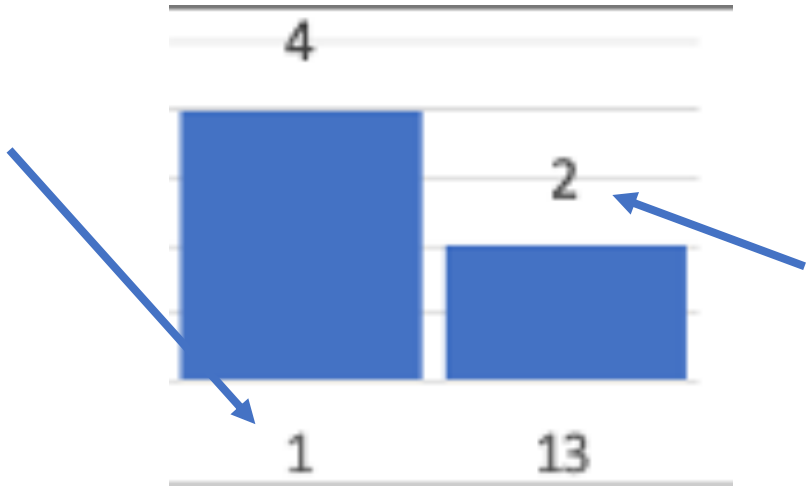


The outcome being examined

Understanding the harvest plot:

- **Each bar represents one paper.**

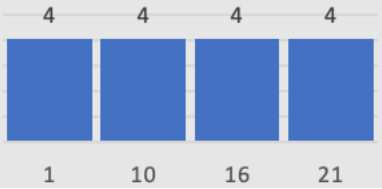
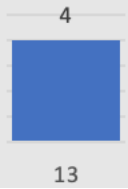
- The number at the base of each bar is a reference to the paper in the bibliography (Appendix 5)



- The number at the top of the bar is the height.
- The height represents the quality of the paper based on if it was downgraded on any of 3 parameters: bias, imprecision, indirectness.

Bias, Imprecision, Indirectness	
	Height
If not graded down	4
If graded down on 1 parameter	3
If graded down on 2 parameters	2
If graded down on 3 parameters	1

Understanding the harvest plot: Example

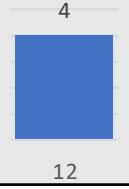
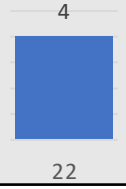
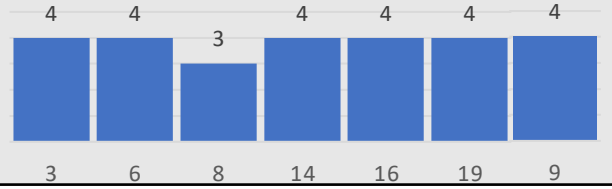
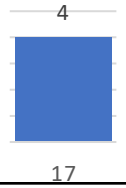
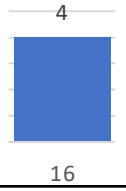
Increased BMI -spinal cord	**Negative correlation**	**No correlation**	**Positive correlation**
Patient-reported outcomes			

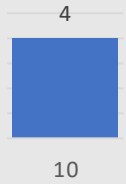
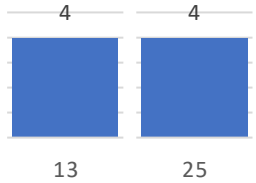
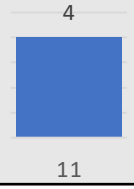
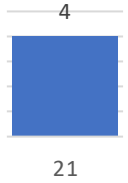
This row shows papers looking at the relationship between increased BMI and patient reported outcomes (under the spinal cord biology category):

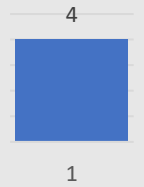
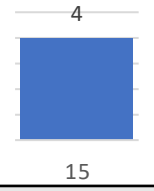
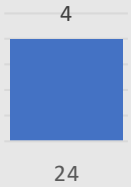
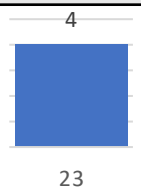
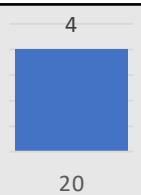
- 4 papers showed a negative correlation between these two variables,
- 1 paper showed no correlation,
- 0 papers showed a positive correlation.

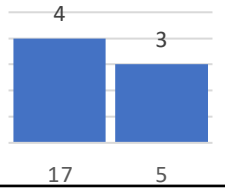
S4. Appendix 4. Harvest Plot: Results

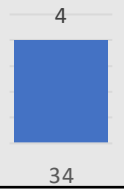
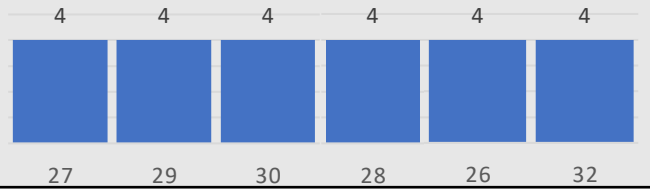
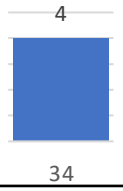
This is a visual representation of the specific outcomes assessed in individual papers (refer to S5.Appendix 5 for bibliography).

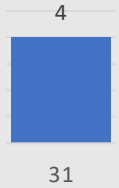
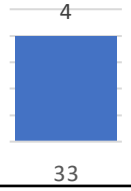
Increased BMI-adverse events	**Negative correlation**	**No correlation**	**Positive correlation**
Post-operative complications			
Mortality			

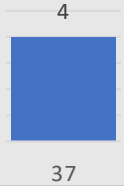
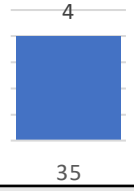
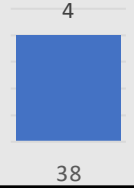
Increased BMI -spinal cord biology	**Negative correlation**	**No correlation**	**Positive correlation**
Patient-reported outcomes			
Change in JOA score			
Functional independence measure efficiency			
Reactive oxygen metabolites			

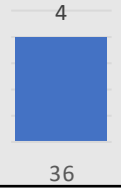
Increased BMI -spinal column biology	**Negative correlation**	**No correlation**	**Positive correlation**
Cervical alignment parameters			
Modic changes			
Loss of cervical lordosis			
Adjacent segment disease			
Risk of cervical spondylosis			

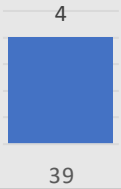
Pre-operative weight loss- adverse events	**Negative correlation**	**No correlation**	**Positive correlation**
Post-operative complications			
Mortality			

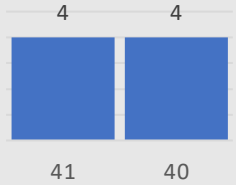
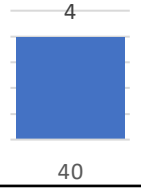
Malnutrition/electrolytes-Adverse events	**Negative correlation**	**No correlation**	**Positive correlation**
Post-operative complications			
Time for fusion			

Malnutrition/electrolytes-Spinal cord biology	**Negative correlation**	**No correlation**	**Positive correlation**
Functional independence measure efficiency			
Change in post-operative JOA score (RCT)			

Vitamins/minerals-spinal cord biology	**Negative correlation**	**No correlation**	**Positive correlation**
Upper extremity work capacity			
Change in post-operative JOA score (RCT)			
Duration of spinal cord injury			

Vitamins/minerals-spinal column biology	**Negative correlation**	**No correlation**	**Positive correlation**
Risk of OPLL development			

Gastrointestinal health-Adverse events	**Negative correlation**	**No correlation**	**Positive correlation**
Post-operative complications			

Gastrointestinal health-Spinal cord biology	**Negative correlation**	**No correlation**	**Positive correlation**
Patient-reported outcomes			
Change in post-operative JOA score